

一类新的KdV方程的周期解

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摘 要

本文根据三波法的思想, 结合一类新的KdV方程的Hirota双线性形式, 通过选择特定形式的试探函数, 构造出该方程的孤波解与周期解, 并对解中的参数赋值, 利用计算机符号软件对部分解进行了数值模拟, 观察解的三维波形图与动力学演化过程。

关键词

KdV方程, Hirota双线性形式, 周期解, 孤波解

A New Class of Periodic Solutions to the KdV Equation

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Abstract

In this paper, based on the idea of the three-wave method, combined with a new class of Hirota bilinear forms of the KdV equation, we constructed the solitary-wave solution and the periodic solution of the equation by choosing a specific form of the trial function and assigning values to the parameters in the solution, and numerically simulated some of the solutions by using the computer symbolic software, observing the three-dimensional waveform plots of the solutions with the dynamics evolution process.

Keywords

KdV Equation, Hirota Bilinear Form, Periodic Solutions, Solitary-Wave Solutions

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1. 引言

KdV 方程

$$u_{xxx} + u_t + 6uu_x = 0, \quad (1)$$

结合了非线性项和色散项,成功解释了孤立波的稳定传播机制,其中 $u = u(x, t)$ 。KdV 方程的各类扩展型模型在流体力学、等离子体物理和非线性光学等领域有广泛的应用, KdV 方程的理论成果不仅推动了数学物理的进步,还启发了光纤通信中的孤子传输、量子场论中的瞬子解等实际应用,体现了基础科学与工程技术的紧密结合[1]-[4]。

在众多 KdV 型模型中,有一类新提出的(2 + 1)维 KdV 方程[5],它可以用来描述非线性波,例如浅水波、地表波和内部波

$$u_{xxy} + u_t + 3u_x u_y = 0, \quad (2)$$

其中, $u = u(x, y, t)$ 。对 KdV 方程不同类型的非线性波解[6]-[8],如孤子解、多孤子解、周期解、呼吸子解的构造不仅可以丰富数学物理的理论体系,还为实际现象的解释和技术应用提供了关键工具,研究其行为有助于揭示复杂系统的动力学机制,提升预测和防范自然灾害的能力。而 KdV 方程的非线性波的模拟又往往难以在实验室中进行,因此构造 KdV 方程的非线性波解就成了热门的研究内容,对于该类 KdV 方程,研究者们已经得到了许多有价值的研究成果[9] [10]。

2. 新的 KdV 方程的周期解

作一个对数变换

$$u = 2(\ln f)_x, \quad (3)$$

可以得到方程(2)的 Hirota 双线性形式

$$(D_t D_x + D_x^3 D_y) f \cdot f = 0, \quad (4)$$

方程(4)等价于

$$ff_{xt} - f_t f_x + 3f_{xy} f_{xx} - 3f_x f_{xxy} - f_y f_{xxx} + ff_{xxy} = 0. \quad (5)$$

根据三波法[11] [12]的思想,假设方程(5)具有如下形式的解

$$f = k_1 \cosh(\xi_1) + k_2 \cosh(\xi_2) + k_3 \cos(\xi_3) + a_0, \quad (6)$$

其中

$$\begin{aligned} \xi_1 &= a_1 x + a_2 y + a_3 t + a_4, \\ \xi_2 &= a_5 x + a_6 y + a_7 t + a_8, \\ \xi_3 &= b_1 x + b_2 y + b_3 t + b_4. \end{aligned} \quad (7)$$

上式中, $a_i (i = 0, 1, \dots, 8)$, $b_i (i = 1, 2, 3, 4)$, $k_i (i = 1, 2, 3)$ 是待定常数。将(6)与(7)式代入方程(5),再令 $\sinh(\xi_1), \cosh(\xi_1), \sinh(\xi_2), \cosh(\xi_2), \sin(\xi_3), \cos(\xi_3)$ 的各次幂系数为零,可以得到由 $a_i (i = 0, 1, \dots, 8)$, $b_i (i = 1, 2, 3, 4)$, $k_i (i = 1, 2, 3)$ 组成的大规模代数方程组,利用符号计算软件 Maple 进行求解,可以得到使代

数方程组成立的 9 组约束关系。

第 1 组

$$\begin{aligned} a_0 = 0, a_1 = a_1, a_2 = -\frac{1}{4} \frac{a_3}{a_1^2}, a_3 = a_3, a_4 = a_4, a_5 = -a_1, a_6 = -\frac{1}{4} \frac{a_7}{a_1^2}, \\ a_7 = a_7, a_8 = a_8, b_1 = b_1, b_2 = b_2, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = 0. \end{aligned} \quad (8)$$

第 2 组

$$\begin{aligned} a_0 = 0, a_1 = a_1, a_2 = -\frac{1}{4} \frac{a_3}{a_1^2}, a_3 = a_3, a_4 = a_4, a_5 = a_1, a_6 = -\frac{1}{4} \frac{a_7}{a_1^2}, \\ a_7 = a_7, a_8 = a_8, b_1 = b_1, b_2 = b_2, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = 0. \end{aligned} \quad (9)$$

第 3 组

$$\begin{aligned} a_0 = a_0, a_1 = a_1, a_2 = 0, a_3 = 0, a_4 = a_4, a_5 = 0, a_6 = -\frac{a_7}{a_1^2}, a_7 = a_7, \\ a_8 = a_8, b_1 = b_1, b_2 = b_2, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = 0. \end{aligned} \quad (10)$$

第 4 组

$$\begin{aligned} a_0 = a_0, a_1 = a_1, a_2 = 0, a_3 = 0, a_4 = a_4, a_5 = 0, a_6 = -\frac{a_7}{a_1^2}, a_7 = a_7, \\ a_8 = a_8, b_1 = b_1, b_2 = b_2, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = 0. \end{aligned} \quad (11)$$

第 5 组

$$\begin{aligned} a_0 = a_0, a_1 = a_1, a_2 = a_2, a_3 = a_3, a_4 = a_4, a_5 = a_5, a_6 = 0, a_7 = 0, \\ a_8 = a_8, b_1 = 0, b_2 = -\frac{b_3}{a_5^2}, b_3 = b_3, b_4 = b_4, k_1 = 0, k_2 = k_2, k_3 = k_3. \end{aligned} \quad (12)$$

第 6 组

$$\begin{aligned} a_0 = a_0, a_1 = a_1, a_2 = a_2, a_3 = a_3, a_4 = a_4, a_5 = a_5, a_6 = 0, a_7 = 0, \\ a_8 = a_8, b_1 = 0, b_2 = -\frac{b_3}{a_5^2}, b_3 = b_3, b_4 = b_4, k_1 = 0, k_2 = k_2, k_3 = k_3. \end{aligned} \quad (13)$$

第 7 组

$$\begin{aligned} a_0 = a_0, a_1 = a_1, a_2 = 0, a_3 = 0, a_4 = a_4, a_5 = 0, a_6 = -\frac{a_7}{a_1^2}, a_7 = a_7, \\ a_8 = a_8, b_1 = 0, b_2 = -\frac{b_3}{a_1^2}, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = k_3. \end{aligned} \quad (14)$$

第 8 组

$$\begin{aligned} a_0 = a_0, a_1 = a_1, a_2 = 0, a_3 = 0, a_4 = a_4, a_5 = -a_1, a_6 = 0, a_7 = 0, \\ a_8 = a_8, b_1 = 0, b_2 = -\frac{b_3}{a_1^2}, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = k_3. \end{aligned} \quad (15)$$

第 9 组

$$a_0 = a_0, a_1 = a_1, a_2 = 0, a_3 = 0, a_4 = a_4, a_5 = a_1, a_6 = 0, a_7 = 0,$$

$$a_8 = a_8, b_1 = 0, b_2 = -\frac{b_3}{a_1^2}, b_3 = b_3, b_4 = b_4, k_1 = k_1, k_2 = k_2, k_3 = k_3. \quad (16)$$

接下来, 以上面 9 组中的部分代数解为例, 给出方程(2)周期解的具体形式, 并进行数值模拟。将(8)式代入(6)与(7)式, 结合变换(3)可得

$$u_1 = \frac{2 \left(k_1 a_1 \sinh \left(a_1 x - \frac{1}{4} \frac{a_3 y}{a_1^2} + a_3 t + a_4 \right) - k_2 a_1 \sinh \left(-a_1 x - \frac{1}{4} \frac{a_7 y}{a_1^2} + a_7 t + a_8 \right) \right)}{k_1 \cosh \left(a_1 x - \frac{1}{4} \frac{a_3 y}{a_1^2} + a_3 t + a_4 \right) + k_2 \cosh \left(-a_1 x - \frac{1}{4} \frac{a_7 y}{a_1^2} + a_7 t + a_8 \right)}, \quad (17)$$

将(12)式代入(6)与(7)式, 结合变换(3)可得

$$u_2 = \frac{2k_2 a_5 \sinh(a_5 x + a_8)}{k_2 \cosh(a_5 x + a_8) + k_3 \cos \left(-\frac{b_3 y}{a_5^2} + b_3 t + b_4 \right) + a_0}, \quad (18)$$

将(14)式代入(6)与(7)式, 结合变换(3)可得

$$u_3 = \frac{2k_1 a_1 \sinh(a_1 x + a_4)}{k_1 \cosh(a_1 x + a_4) + k_2 \cosh \left(-\frac{a_7 y}{a_1^2} + a_7 t + a_8 \right) + k_3 \cos \left(-\frac{b_3 y}{a_1^2} + b_3 t + b_4 \right) + a_0}. \quad (19)$$

在解 u_1 中取参数为 $a_1 = 1, a_3 = 1, a_4 = 1, a_7 = 1, a_8 = 1, k_1 = 1, k_2 = 1$ 及 $t = -20, t = 0, t = 20$, 可以得到方程(2)的纽结孤立波解, 如图 1(a)所示, 在不同时刻, 孤立波解的形状、传播速度均不会随时间发生改变, 具有很好的稳定性, 图 1(b)、图 1(c)还给出了孤立波解的等高线图和二维图。

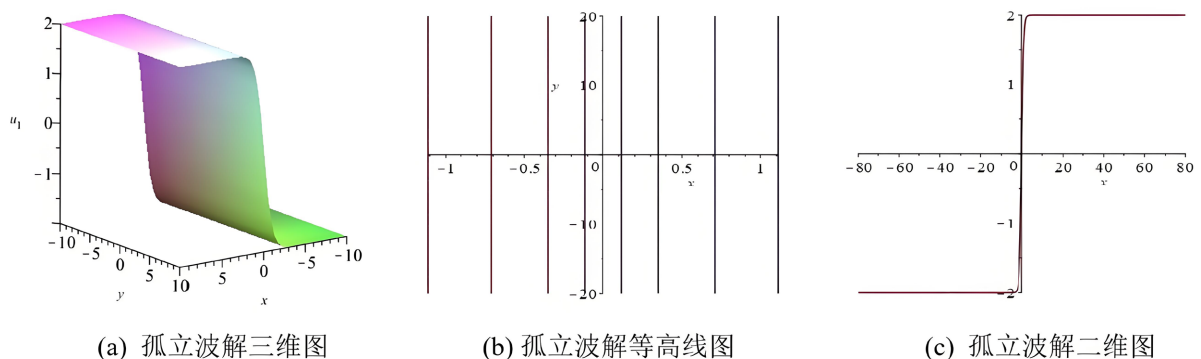


Figure 1. Three-dimensional, contour and two-dimensional maps of u_1

图 1. 解 u_1 的三维图、等高线图与二维图

在解 u_2 中取参数为 $a_0 = 0.2, a_5 = 1, a_8 = 3, b_3 = 1, b_4 = -1, k_2 = 1, k_3 = 2$, 分别在 $t = -20, t = 10, t = 20$ 三个时刻对解 u_2 进行数值模拟, 动力学演化过程如图 2(a)~(c)所示, 展示了孤波与周期波之间的相互作用过程, 可以发现孤波与周期波在相互作用时孤波的结构保持不变, 周期波随着时间的推移沿着 y 轴的正半轴传播。

在解 u_3 中取参数为 $a_0 = 1, a_1 = 2, a_4 = 1, a_7 = 0.3, a_8 = 1, b_3 = -1, b_4 = 1, k_1 = -1, k_2 = 1, k_3 = 1$, 分别在 $t = -20, t = 0, t = 20$ 三个时刻对解 u_3 进行数值模拟, 动力学演化过程如图 3(a)~(c)所示, 展示了两个孤波与周期波之间有趣的相互作用现象。在相互作用过程中, 两个孤波始终保持原有形状与速度不受影响, 周期波在相互作用的过程中沿着 x 轴的正半轴进行传播。

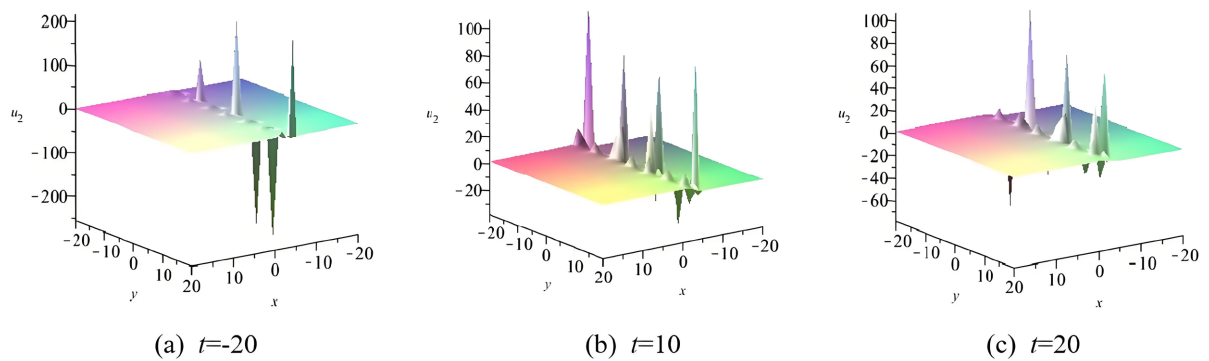


Figure 2. Three-dimensional maps of the kinetic evolution of u_2

图 2. 解 u_2 动力学演化的三维图

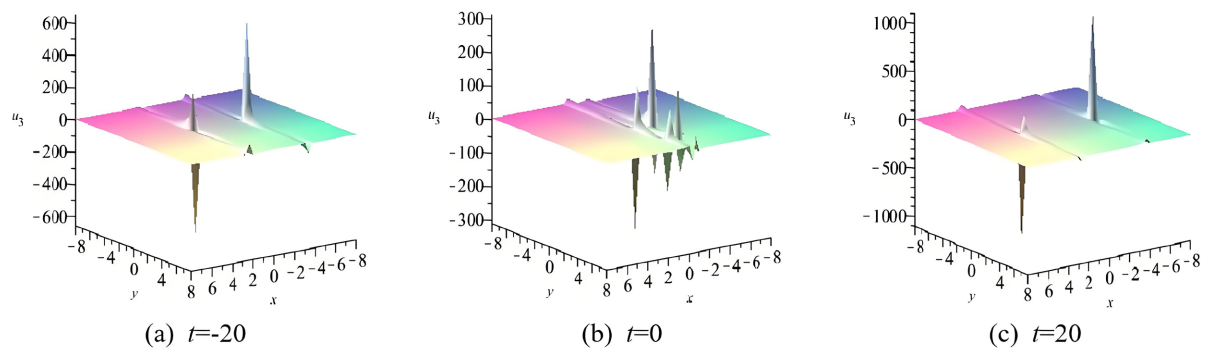


Figure 3. Three-dimensional maps of the kinetic evolution of u_3

图 3. 解 u_3 动力学演化的三维图

3. 结论

本文利用一类新的 KdV 方程的 Hirota 双线性形式, 根据三波法思想, 选择特定形式的试探函数, 构造了新的 KdV 方程的周期解, 并给出了部分解的具体形式。对解中的参数赋值并进行数值模拟, 观察孤波解与周期解的物理结构与动力学行为, 可以发现, 无论是单个孤波还是双孤波, 在与周期波相互作用的过程中始终能够保持原有结构, 具有很好的稳定性, 展示了非线性波的复杂叠加效应, 为研究波与波之间的相互作用、能量传递提供了理论依据和切实方法。

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